

Clinical Data Modeling with SULO

Remzi Celebi, Maastricht University, the Netherlands
Catalina Martínez-Costa, University of Murcia, Spain

Schedule

Start Time	Duration	Session / Topic	Presenter(s)
09:00	00:15	1. SULO WHY	Remzi
09:15	00:20	2. SULO HOW	Cati
09:35	00:55	3. Common clinical concepts (Visit, Lab result, Procedure, Treatment, Surgery, Drug, Condition) - OWL axiom formulation and owlready2 basic implementation	Remzi
10:30	00:30	Break	
11:00	00:30	Use cases and owlready2 implementation (short exercises, slides + collab)	Cati
11:30	01:00	Interactive session and discussion (more complex concepts in Cancer)	Cati and Remzi
13:00		End	

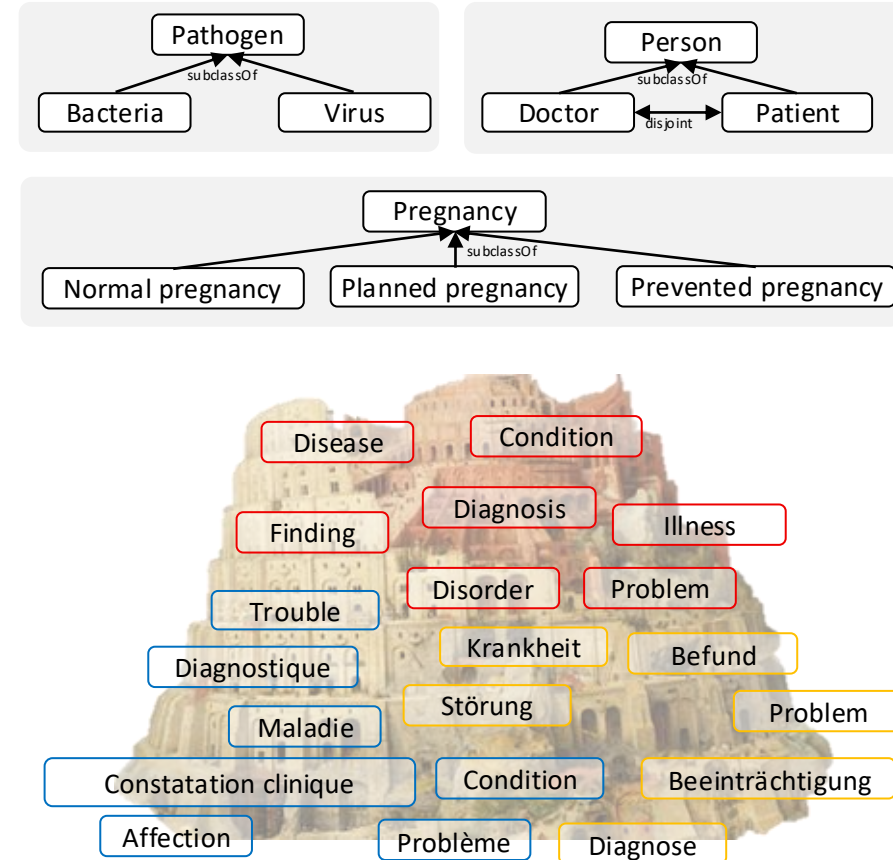


SULO: WHY?

(value prop)

Failure of semantic interoperability in health

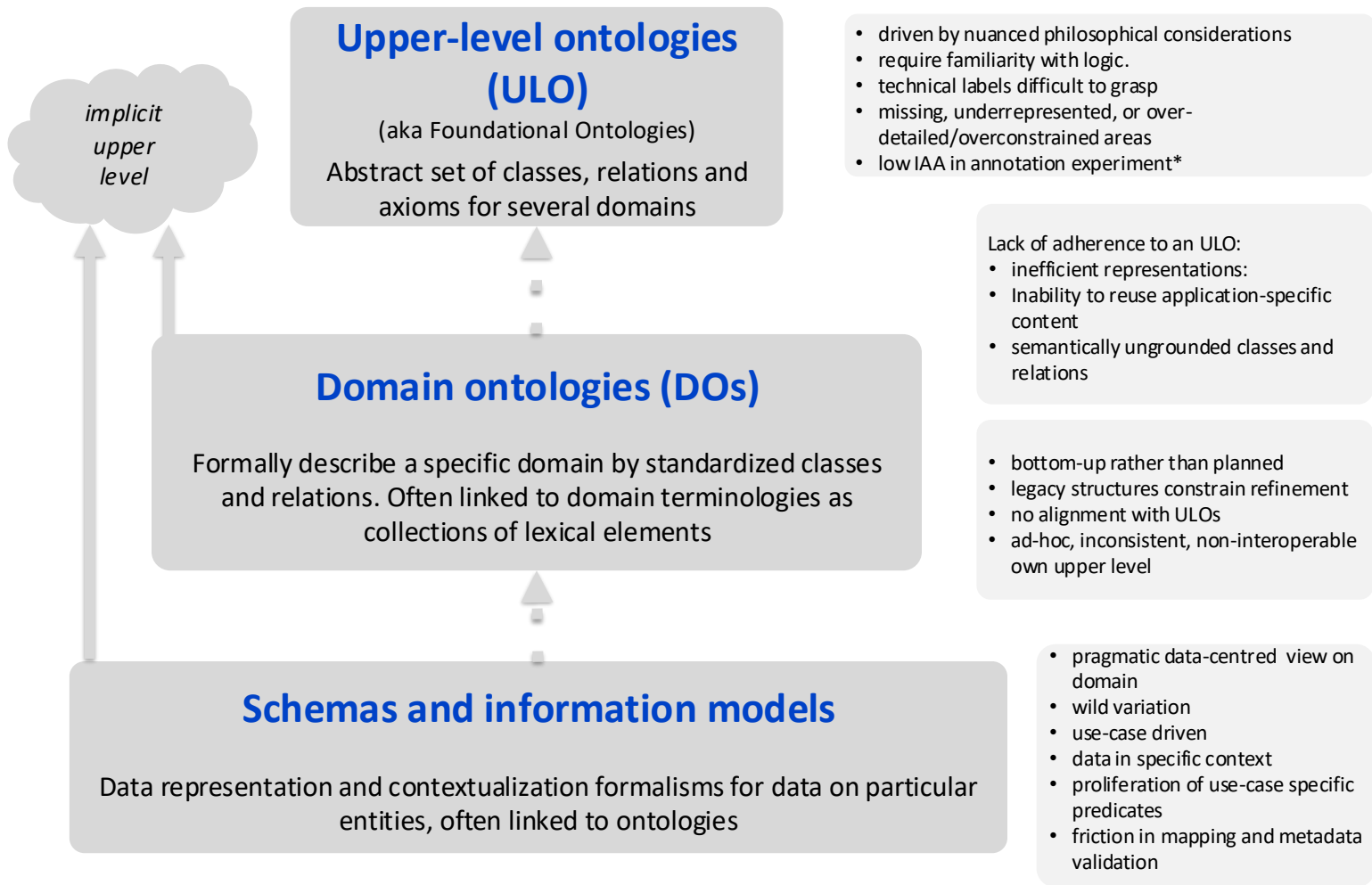
- Not because of technology but of **lack of shared semantics:**
 - bottom-up modelling
 - implicit assumptions, and contexts
 - scope driven by use cases
- Modelling decisions are language-driven
 - Different natural languages, dialects, sociolects
 - No 1:1 translation of general terms
 - Sloppy labelling instead of precise defining
- Upper-level assumptions are **implicit**
 - who rejects the need of a shared reference of meaning that
 - their own, implicit, mental concepts are shared by everybody
 - Every interoperability project reinvents an ontology, and poorly
- Common ontologies as a solution?



Examples.
BFO, DOLCE,
GFO, UFO,
SUMO

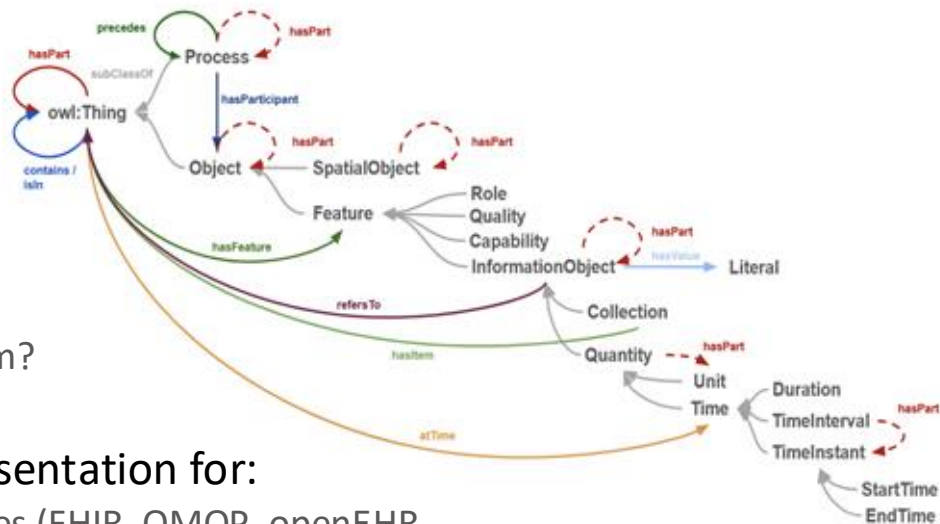
OBO ontologies,
e.g. GO, OBI,
HPO,
openGALEN
SNOMED CT

HL7-FHIR
openEHR
OMOP
SPHINX



Solution: Simple Upper Level Ontology - SULO

- Interoperability requires shared primitives, not shared databases.
- Few clear, reusable upper-level classes and relations
 - What kinds of entities are there?
 - What kinds of relations exist between them?
 - Focus on qualities, roles, quantities, units
- A concise shared ontology-based representation for:
 - Models / schemas as information ontologies (FHIR, OMOP, openEHR, SPHN, etc.)
 - Neurosymbolic pipelines to support semantic enrichment for graph embeddings and KG completion
 - Automated, reproducible FAIR semantic engineering ecosystems



Challenges

Existing ontologies are too heavy, too inconsistent, or too domain-specific.

- Foundational ontologies (BFO, DOLCE) are powerful but **too abstract** and difficult to operationalize for applied health data engineering.
- Domain terminologies (SNOMED, LOINC) are vast, highly specialised, and **not designed for conceptual normalisation across heterogeneous data sources**.
- **Project-level data models vary wildly**, leading to high friction in mapping, transformation, and metadata validation.

Result: Every interoperability project reinvents an ontology, and poorly.

Solution: SULO offers a principled, flexible backbone for ontology engineering efforts.

SULO offers a sufficient minimal semantics

Real-world healthcare data needs just enough semantics, not maximal metaphysics.

Hospital data engineers, FAIR stewards, and ML teams need:

- A small set of clear, reusable upper-level distinctions (process vs object; spatial object vs information object; role vs capability).
- Effective patterns, not philosophical debates.
- Predictable modeling choices so that tooling - ETL, quality checks, mappings, agents - can automate more and guess less.

SULO provides a deliberately small, practical, and friendly upper-level ontology.

SULO solves a fundamental problem in data interoperability

SULO solves a fundamental problem: **Interoperability requires shared modeling primitives, not shared databases.**

SULO gives projects a common blueprint for:

- What kinds of entities are there?
- What kinds of relations exist between them?
- How to represent qualities, quantities, roles, and measurements and their units

This means:

- FHIR resources, OMOP tables, cohort datasets, registries, and device data can all be lifted into a shared conceptual representation.
- SHACL, SheXMap, and SPARQL transformations become massively simpler and safer.
- LLM-based metadata augmentation becomes more reliable because the ontology enforces consistent patterns



SULO: How?

SULO

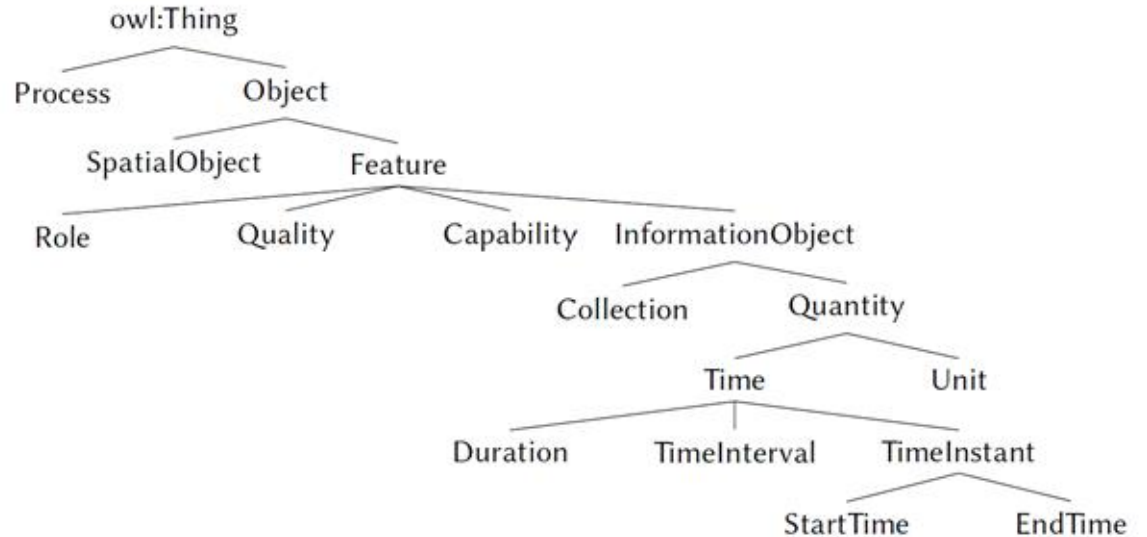
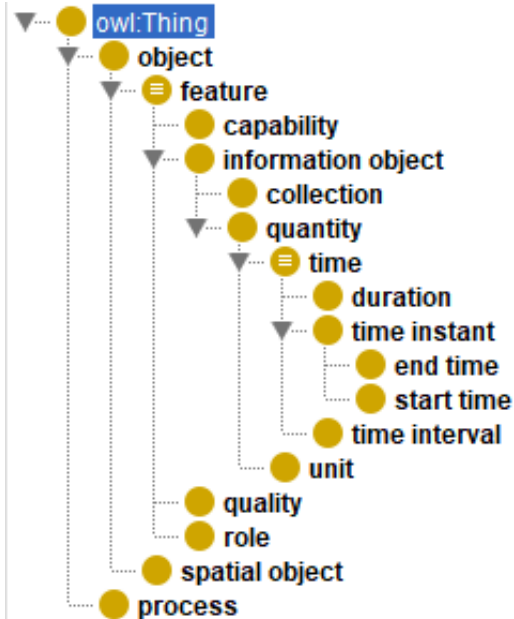
We propose the **Simplified Upper Level Ontology (SULO)** that takes a minimalistic approach to guide the alignment, formalisation, and reusability of upper level and domain ontologies.

SULO attempts to **balance formal rigour with simplicity and practical usability**.

Design Criteria

- **Minimalism:** SULO proposes a small taxonomy of disjoint classes and a minimal set of constrained relations to ensure broad applicability across domains.
- **Compatibility:** SULO maintains compatibility with core components of well-known ULOs while remaining accessible to domain experts.
- **Accessibility:** SULO aims to be accessible to users with no or little training in formal ontology through friendly labeling, a simple taxonomy, and fits in a single diagram.
- **Composability:** SULO provides the building blocks to construct complex, machine-readable class expressions.
- **Interoperability:** SULO fosters interoperability by providing a common semantic foundation, including two ontology design patterns, that help domain experts adhere to explicit and implicit semantics.
- **Data validation:** SULO constrains real-world knowledge graphs through automated reasoning.

17 classes in SULO



Process and Object



Object. An object is a thing that exists and keeps its identity over time, even if it changes a little. It has parts and features, and you can usually point to it or describe it as something. **Examples:** a pizza, a heart, the role of being a health care provider, the colour of an eye, a document, a plan.

If it can be described as something that persists through time, it is an object.

Process. A process is a thing that occurs over time. It has temporal parts (e.g. a beginning, middle, and end), and it changes or moves objects. **Examples:** Baking a pizza, the examination of a patient, the administration of medication, running a computer program.

If it unfolds over time and involves interaction, motion, change, or

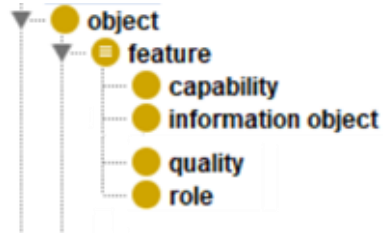
Spatial Object and Feature



Spatial Object: A spatial object is an Object with a physical extension in space. It exists independently of other things, can have parts, and participates in processes. **Examples:** A heart, a pizza, the pepperoni topping on a pizza, a sample of blood.

Feature: A Feature is an Object that *depends on* another thing for its existence. It can describe, qualify, enable, or represent aspects of an object or process. **Examples:** The color of a heart, the capability to pump blood, the temperature of the air, a blog post about heart pumping.

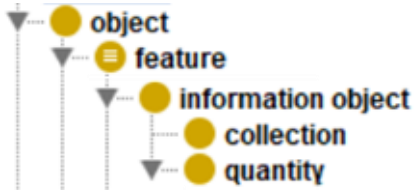
Features



Types of features include:

- **Quality:** characteristics (color, mass, height)
- **Capability:** what an object can do under the right conditions (pump blood, transmit information using WiFi)
- **Role:** contextually assigned functions or social/organizational statuses (patient role, reagent role)
- **InformationObject:** symbolic or descriptive content about something (a measurement value, a document, a genotype sequence)

Information Objects

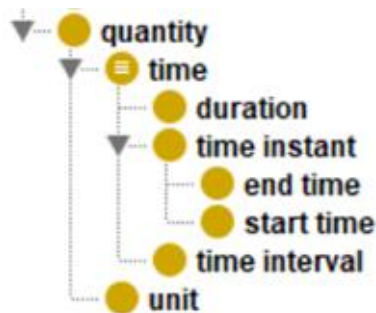


Information Object: An Information Object is anything that carries or expresses information. It may take the form of text, numbers, symbols, codes, images, or structured data. They depend on some physical or digital medium, they are not the physical carriers (e.g. paper). **Examples:** A patient record, a measurement value, a SNOMED-CT code, a data format.

Collection: A collection is an information object that is composed of a set of items (e.g. a set of patient identifiers, a list of medications). Collections do not imply physical grouping.

Quantity: A Quantity is an information object that expresses a measured or computed value. It may include a unit or other descriptors as parts. Examples: Heart rate of 72 +/- 3 bpm, AUC of 0.92.

Time



Time: Time is the way we describe when something happens and how long it lasts.

A **TimeInstant** is a specific moment, like '2025-07-26 at 14:03'.

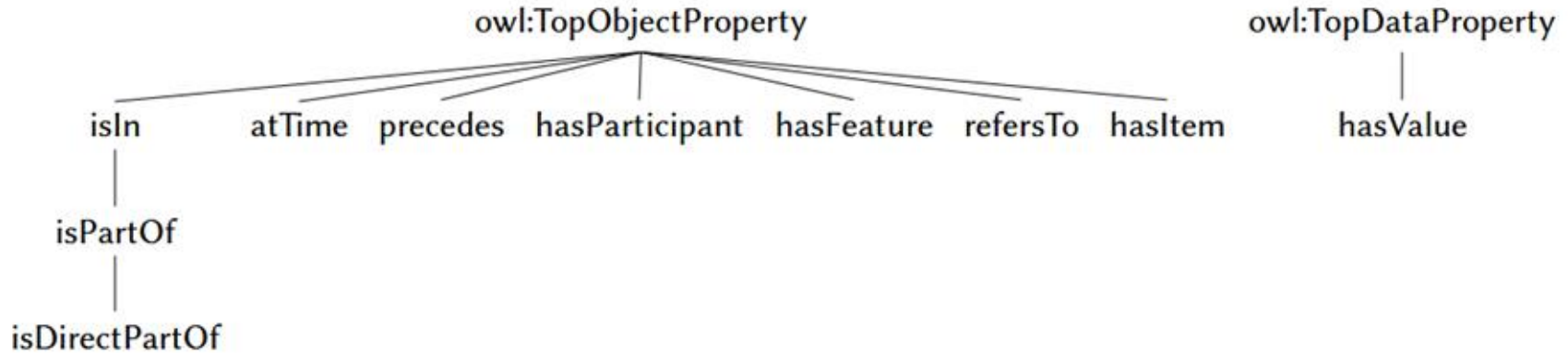
A **TimeInterval** is a span of time that has a beginning and an end.

A **Duration** is a length of time, like 10 seconds, 3 hours, or 6 months.

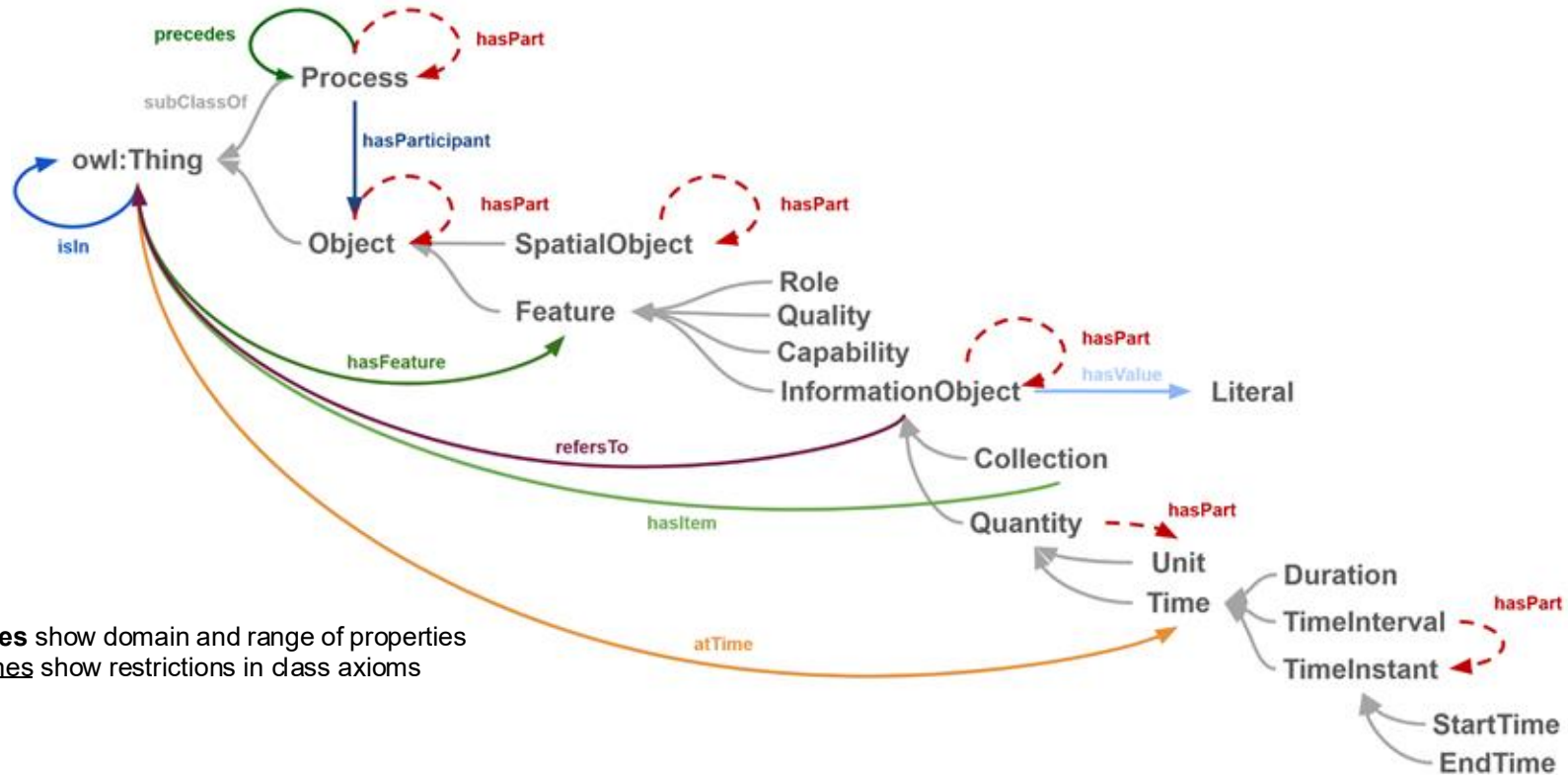
A **Unit** is an InformationObject that specifies the measurement scale or reference frame associated with a numerical value. For instance, the *second* is a unit to measure time.

SULO Properties

18 object properties (9 direct + 9 inverses) and 1 data property.



SULO in a postcard



Ontology Design Patterns

Ontology Design Patterns (ODPs) can help to structure data graphs that foster semantic interoperability across domain-specific knowledge representations

We propose two key ODPs: **SOLID** and **PRO**

SOLID focuses describing *literal* containing relations

PRO focuses on describing *role-based* relations

SOLID ODP

The SOLID pattern uses SULO's single *functional* data property, **hasValue**, to assign a literal value to an InformationObject.

For instance, predicates like **hasTemperature** or **hasTemperatureInCelcius** are reformulated using SULO's **hasValue**, **hasFeature**, **refersTo**, and **hasPart** properties in conjunction with two externally defined classes, namely **Temperature** from PATO and **Celcius** from the Unit Ontology

```
@prefix sulo: <http://w3id.org/sulo/> .
@prefix uo: <http://purl.obolibrary.org/obo/> .
@prefix pato: <http://purl.obolibrary.org/obo/>
@prefix : <http://example.org/> .

:alice a sulo:SpatialObject, :Person;
      sulo:hasFeature :alice_temperature_measurement_1 .

:alice_temperature_measurement_1 a sulo:Quantity;
      sulo:hasValue "37.8"^^xsd:double ;
      sulo:refersTo [ a pato:PATO_0000146 ] ; # the quality of temperature
      sulo:hasPart [ a uo:UO_0000027 ] . # the celcius unit
```

Hence, developing an ontology of InformationObjects is encouraged, rather than a proliferating data properties.

PRO ODP

The Process-Role-Object (PRO) ODP provides a way to represent *the manner in which* **objects** participate in **processes** through specified **roles**.

EquivalentTo:

Process and

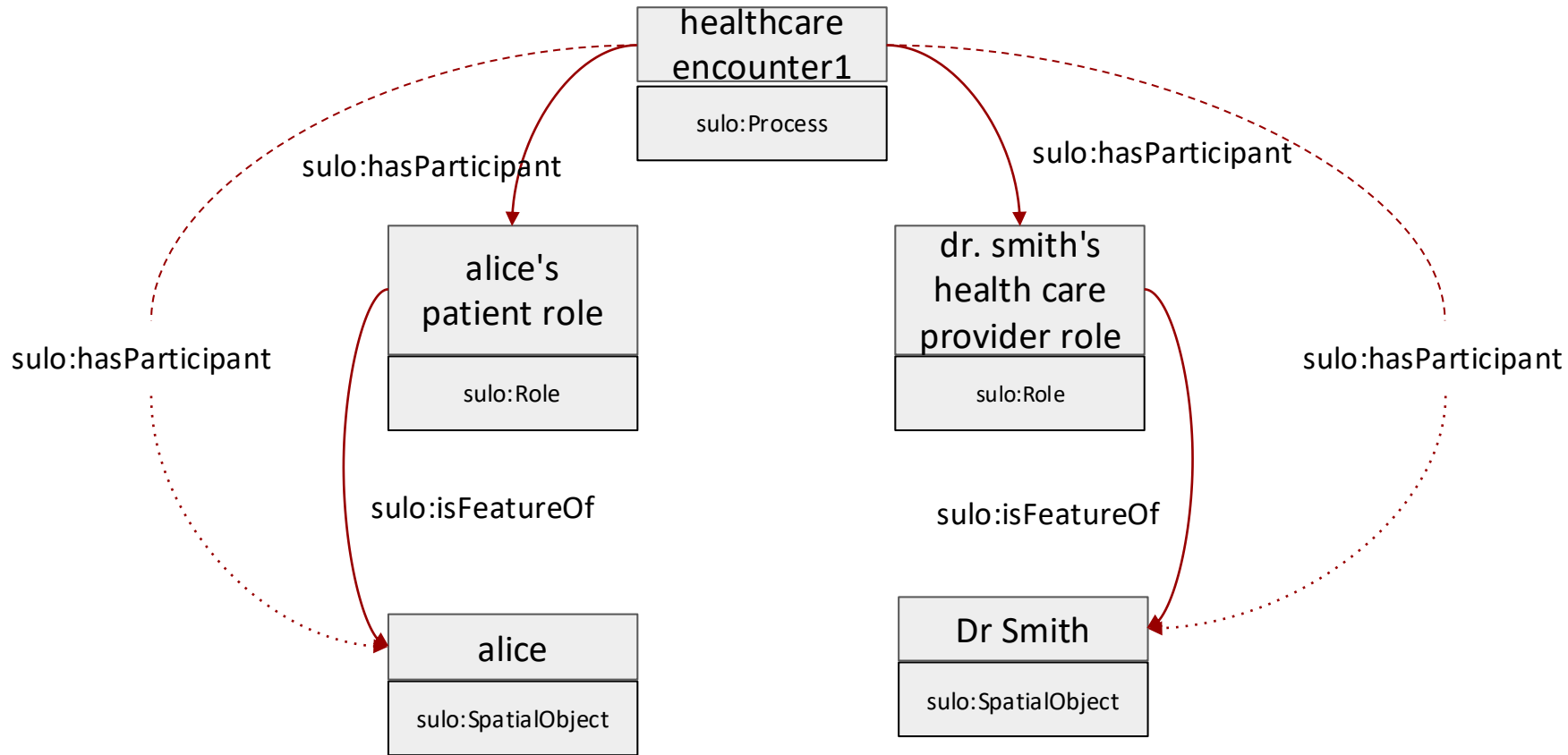
```
hasParticipant some (
  Role and isFeatureOf some Object
)
```

Applies to spatial objects and information objects!

`hasParticipant o isFeatureOf -> hasParticipant`

Role chain to infer participation of role holding object

The pattern makes explicit the semantics for role-aligned object relations such as **hasPatient**, **hasCareProvider**. *Hence, this pattern reduces the impulse to create and proliferate such relations.*



hasParticipant o isFeatureOf -> hasParticipant

SNOMED axioms can be expressed with domain-independent SULO relations

```
Class: 'Fracture of femur (disorder)'  
  
EquivalentTo:  
  'Disease (disorder)'  
  and 'Role group (attribute)' some (  
    'Associated morphology (attribute)' some 'Fracture (morphologic abnormality)'  
    and  
    'Finding site (attribute)' some 'Bone structure of femur (body structure)'  
  )
```

Figure 5: SNOMED class definition for Fracture of Femur (disorder)

```
Class: 'Fracture of femur (disorder)'  
  
EquivalentTo:  
  'Disease (disorder)'  
  and sulo:hasPart some (  
    (sulo:Process and sulo:isIn some  
    'Bone structure of femur (body structure)'  
    and sulo:hasFeature some 'Fracture (morphologic abnormality)')  
  )
```

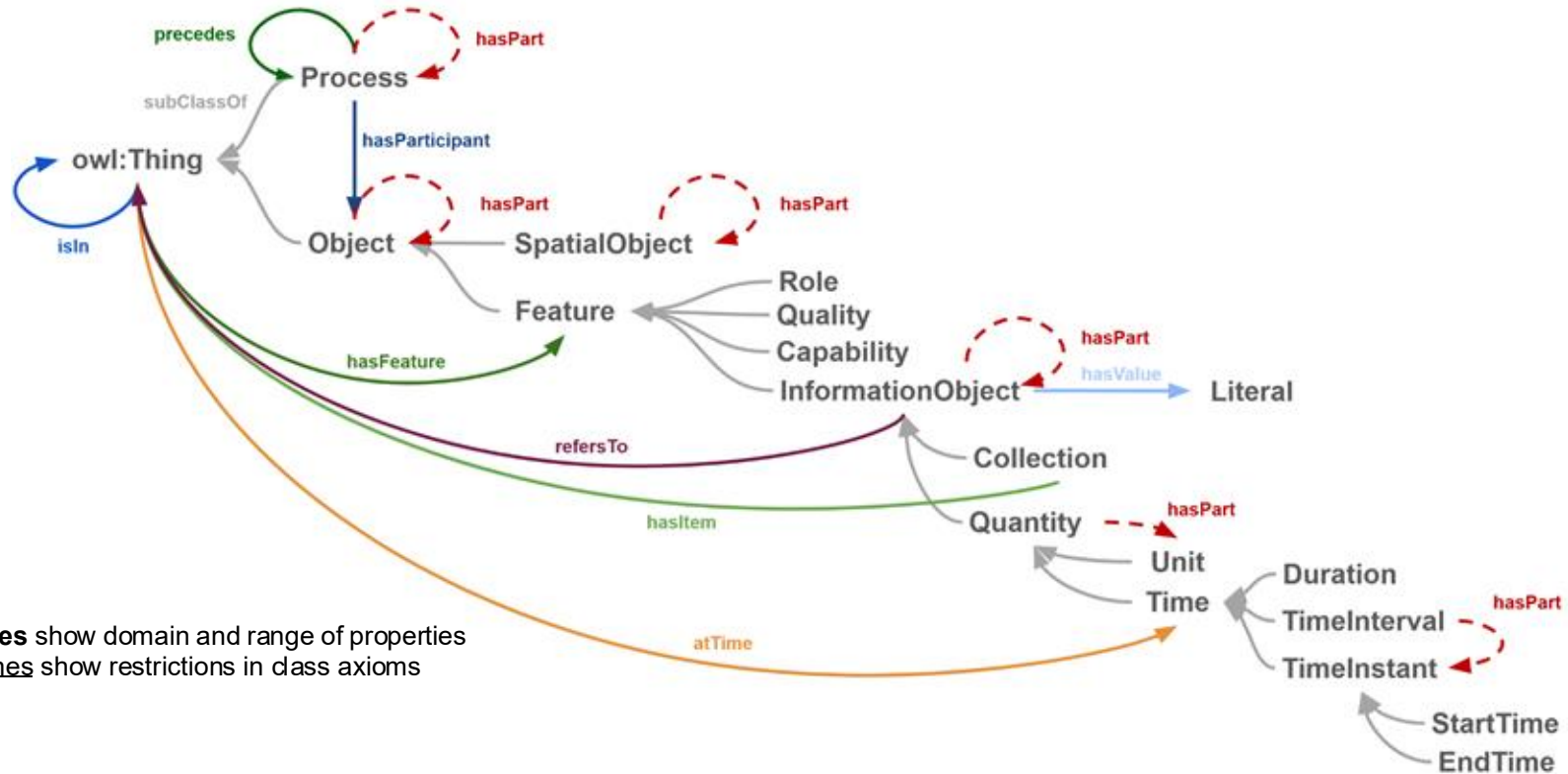
Figure 6: SULO representation of SNOMED Fracture of femur

SULO compared to other ULOs

Fewer classes, fewer relations

Slimmed down, but compatible upper level ontology

SULO in a postcard



solid lines show domain and range of properties
dotted lines show restrictions in class axioms

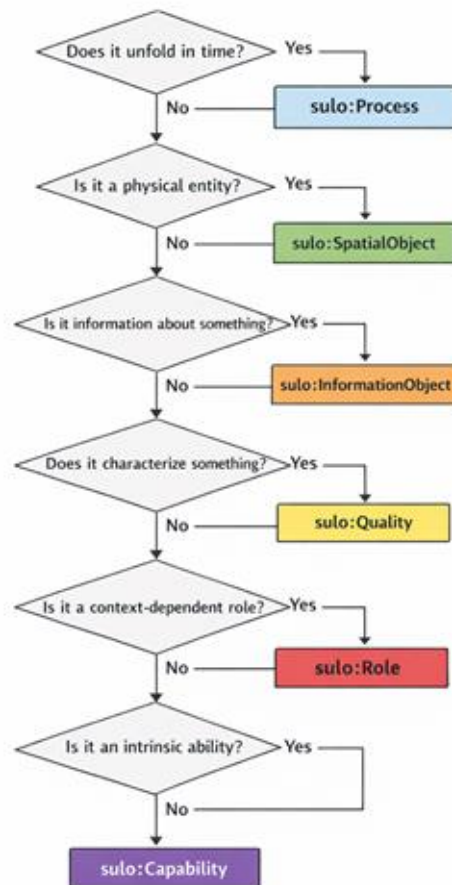
Modeling Healthcare Concepts

Some common concepts mentioned in clinical settings:

- Clinical Visit
- Lab result
- Medical Procedure
- Clinical Condition
- Severity
- Treatment (Drug Administration) / Treatment Plan
- Drug (Pharmaceutical Product)

Flow chart

SULO Classification Decision Flowchart



Step 1: classify the concepts under SULO

- **Clinical Visit: Process**
 - Has a start and end time
 - Has a duration
 - Includes sub-events (check-in, consultation, examination, discharge)
 - Can precede or follow other processes
- **Lab result: InformationObject/Quantity**
 - Is information produced by performing a lab test
 - Refers to a measured attribute (quality) (e.g., glucose level)
 - Has a value (e.g., 6.2 mmol/L) and may include units
 - Is typically recorded in an information system

Step 1: classify the concepts under SULO

- **Medical Procedure: Process**

- A healthcare procedure (e.g., surgery, biopsy, intubation):
- Happens over a time interval
- Has start and end time
- May have sub-processes (anesthesia, incision, suturing)
- May be located in a body site
- May precede recovery or discharge

- **Treatment: Process**

- Happens over time (e.g., chemotherapy cycle, physiotherapy session, medication course)
- Has a start and end
- May consist of multiple sub-processes
- Can be located in a body site
- Can precede recovery

Step 1: classify the concepts under SULO

- **ClinicalCondition: Process**

- Exists over a time interval
- Has a beginning and possibly an end
- Can worsen or improve
- Has phases (acute, chronic, remission)
- Involves a body structure as participant
- May include sub-processes (inflammation, tissue damage)

- **Severity: Quality**

- A degree or intensity of some other entity
- Typically of a clinical conditions such as disease, symptom, injury
- Dependent on that entity

Step 1: classify the concepts under SULO

- **Medication Plan: InformationObject**
 - Refers to some Medication Process
- **Drug: SpatialObject**
 - Is a material entity ((e.g., tablet, vial, infusion bag):).
 - Exists in space.
 - Has mass, shape, composition.
 - Participates in processes (administration, metabolism).

Step 2: Axiomatization

Class: :ClinicalVisit

SubClassOf:

sulo:Process

and sullo:hasParticipant some (:SubjectOfCareRole and sullo:isFeatureOf some :Person)

and sullo:hasParticipant some (:CareProviderRole and sullo:isFeatureOf some :Person)

SubClassOf: sullo:isIn some (:CareUnit)

Step 2: Axiomatization

Class: :LabResult or Measurement

SubClassOf: sulo:Quantity

SubClassOf: sulo:hasValue some xsd:Double

SubClassOf: sulo:hasPart some sulo:Unit

SubClassOf: sulo:atTime some sulo:TimeInstant

SubClassOf: sulo:refersTo some (sulo:Quality)

Step 2: Axiomatization

Class: :MedicalProcedure

SubClassOf: sulo:Process

SubClassOf: sulo:hasParticipant some :(**SubjectOfCareRole** and sulo:isFeatureOf :**Person**)

SubClassOf: sulo:hasParticipant some (: **PerformerRole** and sulo:isFeatureOf :**Person**)

SubClassOf: sulo:hasParticipant some (: **InstrumentRole** and sulo:isFeatureOf :**Device**)

SubClassOf: sulo:hasParticipant some (: **BodySite**)

SubClassOf: sulo:hasFeature some (: **ProcessStatus**)

Step 2: Axiomatization

Class: :MeasurementProcedure

SubClassOf: sulo:Process

SubClassOf: sulo:hasParticipant some (:**OutputRole** and sulo:isFeatureOf InformationObject)

SubClassOf: sulo:hasParticipant some :(**SubjectOfCareRole** and sulo:isFeatureOf :**Person**)

SubClassOf: sulo:hasParticipant some (:**PerformerRole** and sulo:isFeatureOf :**Person**)

SubClassOf: sulo:hasParticipant some (:**InstrumentRole** and sulo:isFeatureOf :**Device**)

SubClassOf: sulo:hasParticipant some (:**BodySite**)

SubClassOf: sulo:hasFeature some (:**ProcessStatus**)

Step 2: Axiomatization

Class: :Condition

SubClassOf: sulo:Process

SubClassOf: sulo:hasParticipant some (:**BodySite**)

SubClassOf: sulo:hasParticipant some (:**OutputRole** and sulo:**isFeatureOf**
:LabResult)

SubClassOf: sulo:hasFeatue some (:**Severity** and sulo:Quality)

Class: :**Severity**

SubClassOf: sulo:Quality

Step 2: Axiomatization

Class: Treatment / MedicationAdministration

SubClassOf: sulo:Process and sulo:MedicalProcedure

SubClassOf: sulo:hasParticipant some :**SubjectOfCareRole**

SubClassOf: sulo:hasParticipant some :**BodySite**

SubClassOf: sulo:hasParticipant some :**PharmaceuticalProduct**

SubClassOf: sulo:hasParticipant some :**ProcessStatus**

SubClassOf: sulo:atTime some (sulo:StartTime) and sulo:atTime some (sulo:EndTime)

Step 2: Axiomatization

Class: :PharmaceutialProduct

SubClassOf: sulo:SpatialObject

SubClassOf: sulo:hasFeature some (:**PharmaceuticalDoseForm** and **sulo:Quality**)

SubClassOf: sulo:hasPart some (:**Substance** and sulo:SpatialObject)

SubClassOf: sulo:hasFeature some (**PharmaceuticalDose** and **sulo:Quantity**)

Step 3: Define missing concepts or Redefine inconsistent concepts

SubjectOfCareRole -> <http://snomed.info/id/410604004>

CareProviderRole -> <http://snomed.info/id/158965000>

Person -> <http://snomed.info/id/125676002>

BodySite -> <http://snomed.info/id/123037004>

PerformerRole -> <http://snomed.info/id/420158005>

ProcessStatus -> <http://snomed.info/id/263490005>

PharmaceuticalDose

OutputRole

InstrumentRole

Step 3: Define missing concepts or Redefine inconsistent concepts

Class: :PharmaceuticalDose

SubClassOf: sulo:Quantity

SubClassOf: sulo:hasValue some xsd:decimal

SubClassOf: sulo:hasPart some sulo:Unit

Class: :InstrumentRole

SubClassOf: sulo:Role

SubClassOf: sulo:isFeatureOf some :Device

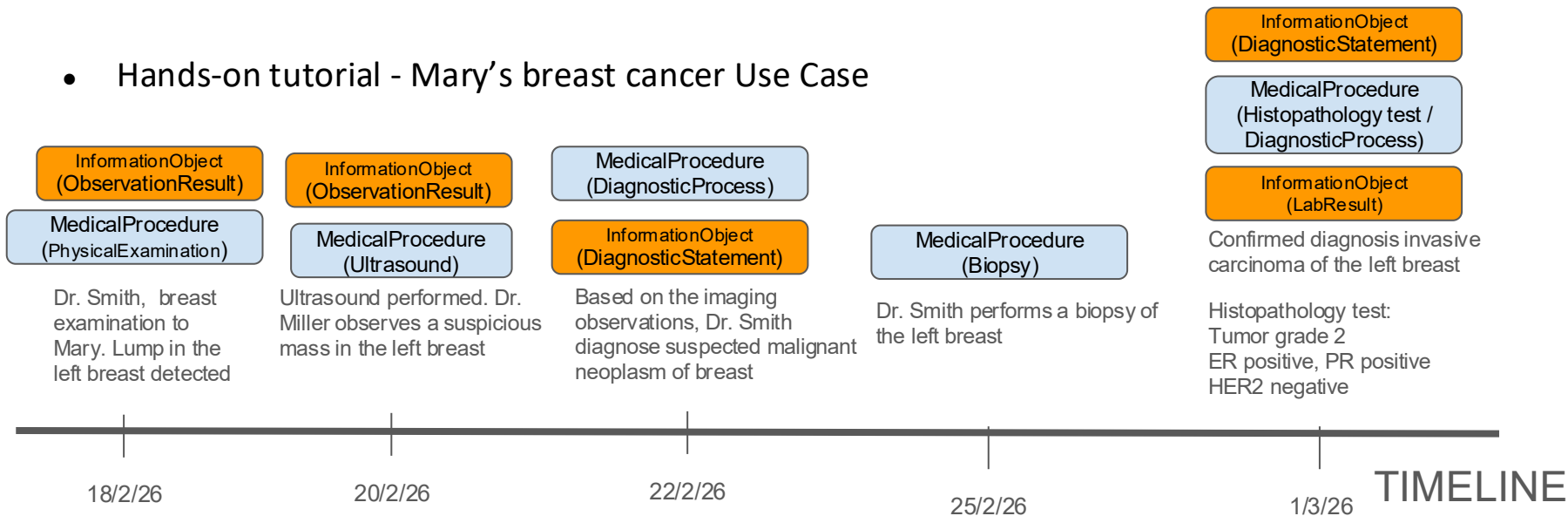
Class: :OutputRole

SubClassOf: sulo:Role

SubClassOf: sulo:isFeatureOf some sulo:InformationObject

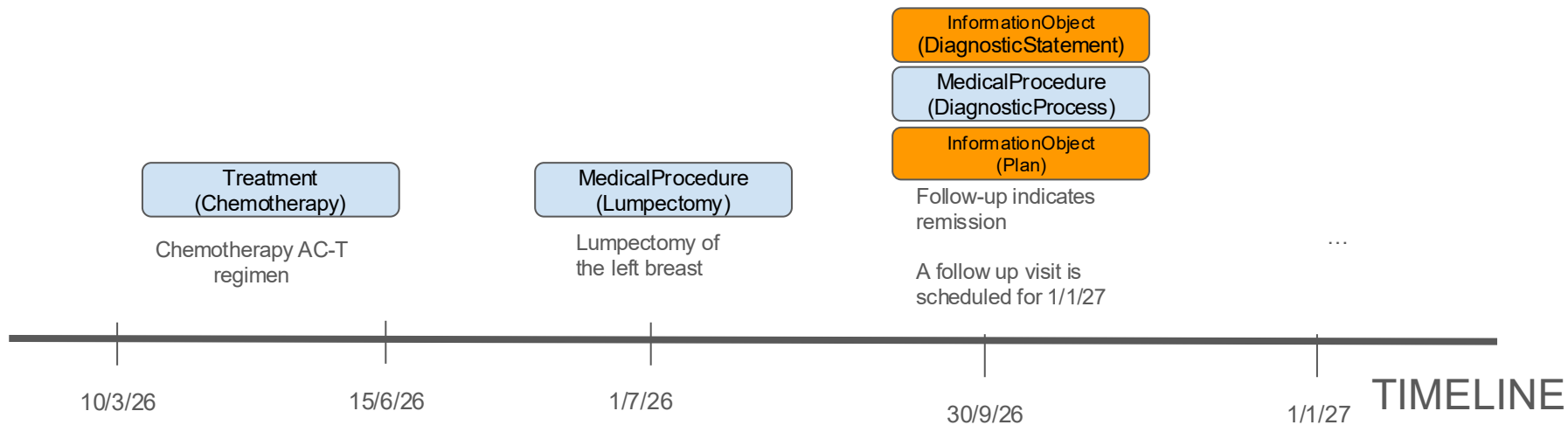
Healthcare Data Modelling Tutorial

- Hands-on tutorial - Mary's breast cancer Use Case



Healthcare Data Modelling Tutorial

- Hands-on tutorial - Mary's breast cancer Use Case



Healthcare Data Modelling Tutorial - Instructions

1. Use SNOMED CT ontology classes whenever possible (named as “SCT_”)
1. Align SNOMED CT classes with SULO upper-level ones
1. Extend SULO only if needed and following its constraints
1. Create the necessary ontology instances for representing the use case data

Healthcare Data Modelling Tutorial - Owlready2

- [Python library for ontology-oriented programming](#)
 - Enables the creation and manipulation of OWL ontologies

- Defining a **Subclass** (green):

```
class SCT_DiagnosticAssessment(sulo.Process):  
    pass
```

- Creating and **Instance** (blue):

```
diag_proc_1 = SCT_DiagnosticAssessment("diag_proc_1")
```

- ☐ Creates an individual of the class
- ☐ Name: "diag_proc_1"

- **Adding Information (rdfs:comment):**

```
diag_proc_1.comment.append("Diagnostic process")
```

- **Adding relations**

```
diag_proc_1.hasParticipant = [pat_mary_role, gyn_smith_role]
```

Healthcare Data Modelling Tutorial - Naming conventions

- SNOMED CT classes: `SCT_<ClinicalConcept>`
 - `SCT_DiagnosticAssessment`
 - `SCT_TumorFinding`
- Individuals: `<abbreviation>_<type>_<index>`
 - `man_bre_exa_proc_1` : manual breast examination process
 - `man_bre_exa_tinst_1` : time instant of the examination

TYPE ABBREVIATIONS	
<code>_proc</code>	Process
<code>_tinst</code>	TimeInstant
<code>_spo</code>	SpatialObject
<code>_io</code>	InformationObject
<code>_role</code>	Role
<code>_qual</code>	Quality

#1 Medical Procedure & Observation result

On 2026-02-18, the patient Mary presented to the outpatient gynecology clinic of Dr. Smith for a routine examination. A physical examination was performed, including a breast examination. On palpation, a breast lump was detected in the left breast.

1. Use SNOMED CT ontology classes whenever possible

a. **SCT_Patient:**

116154003 | Patient (**person**) |

a. **SCT_HospitalBasedOutpatientGynecologyClinic:**

22549003 | Hospital-based outpatient gynecology clinic (**environment**) |

a. **SCT_ManualBreastExamination:**

13607009 | Manual examination of breast (**procedure**) |

a. **SCT_RoutineGynecologicExamination:**

702601001 | Routine gynecologic examination (**procedure**) |

a. **SCT_BreastLump:**

89164003 | Breast lump (**finding**) |

a. **SCT_LeftBreast:**

80248007 | Left breast structure (**body structure**) |

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Manual breast exam (process)
-  Clinic (location)
-  17 Time instant
-  Plan (reason)
-  Observation (breast lump)

#1 Medical Procedure & Observation result

1. Align SNOMED CT classes with SULO upper-level ones

sulo:SpatialObject

a. **SCT_Patient:**

116154003 | Patient (**person**) |

sulo:SpatialObject

a. **SCT_HospitalBasedOutpatientGynecologyClinic:**

22549003 | Hospital-based outpatient gynecology clinic (**environment**) |

sulo:Process

a. **SCT_ManualBreastExamination:**

13607009 | Manual examination of breast (**procedure**) |

sulo:Process

a. **SCT_RoutineGynecologicExamination:**

702601001 | Routine gynecologic examination (**procedure**) |

sulo:Process

a. **SCT_BreastLump:**

89164003 | Breast lump (**finding**) |

sulo:SpatialObject

a. **SCT_LeftBreast:**

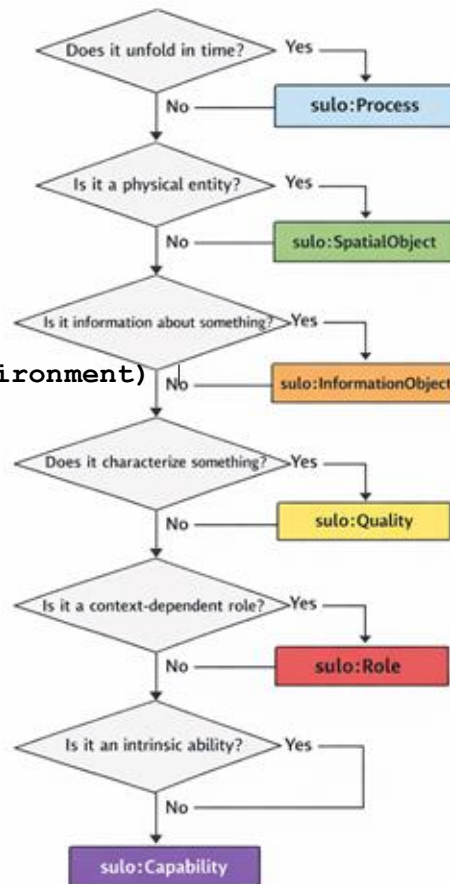
80248007 | Left breast structure (**body structure**) |

sulo:Quality

a. **SCT_Planned:**

397943006 | Planned (**qualifier value**) |

SULO Classification Decision Flowchart



#1 Medical Procedure & Observation result

We need to represent the Person!!

sulo:Role

a. **SCT_Gynecologist:**

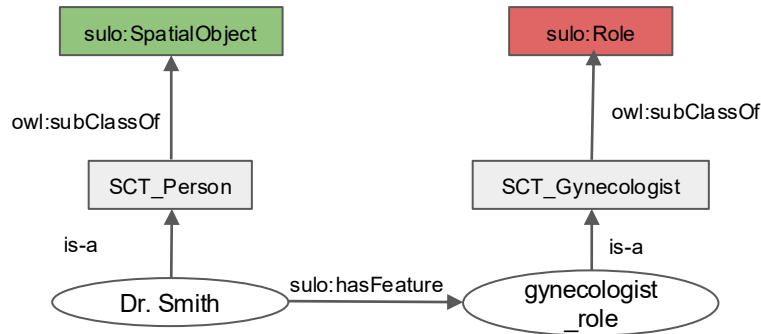
83685006 | Gynecologist (**occupation**) |

sulo:SpatialObject

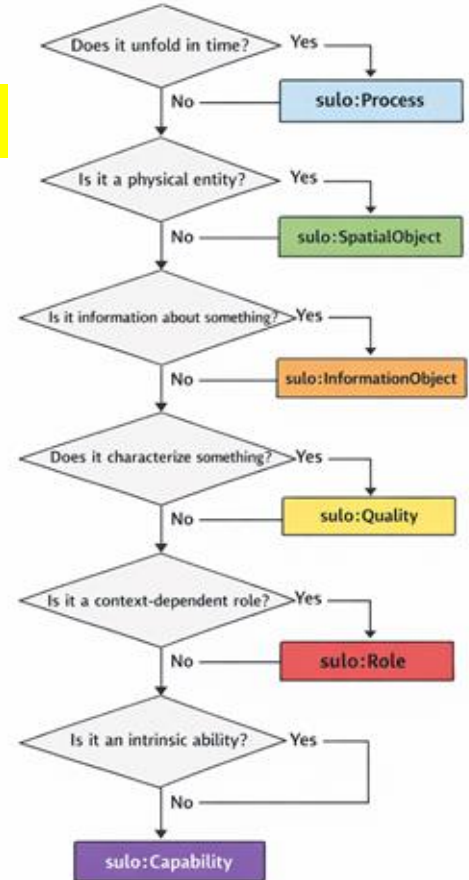
a. **SCT_Person:**

125676002 | Person (**person**) |

“Dr. Smith is a person with the gynecologist role”



SULO Classification Decision Flowchart



#1 Medical Procedure & Observation result

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Manual breast exam (process)
-  Clinic (location)
-  Time instant
-  Plan (reason)
-  Observation (breast lump)

1. Create the necessary ontology instances for representing the use case

```
# instance of manual breast examination process
man_bre_exa_proc_1 = SCT_ManualBreastExamination("man_bre_exa_proc_1")
man_bre_exa_proc_1.comment.append("Manual breast examination process")

# instance of patient Mary
pat_mary_spo = SCT_Person("pat_mary_spo")
pat_mary_role = sulo.Role("pat_mary_role")
pat_mary_spo.hasFeature = [pat_mary_role]

# instance of gynecologist Dr. Smith
gyn_smith_spo = SCT_Person("gyn_smith_spo")
gyn_smith_role = SCT_Gynecologist("smith_gyn_role")
gyn_smith_spo.hasFeature = [gyn_smith_role]

# Both, the gynecologist and the patient are process participants
man_bre_exa_proc_1.hasParticipant = [pat_mary_role, gyn_smith_role]
```

[Link to notebook #1](#)

#1 Medical Procedure & Observation result

```
# time of the examination
man_bre_exa_tinst_1 = sulo.TimeInstant("man_bre_exa_tinst_1")
man_bre_exa_tinst_1.hasValue = datetime.datetime(2026,2,18,0,0)
man_bre_exa_proc_1.atTime = [man_bre_exa_tinst_1]

# the physical location where the examination happened
hosp_out_gyn_cli_spo_1 =
    SCT_HospitalBasedOutpatientGynecologyClinic("hosp_out_gyn_cli_spo_1")
man_bre_exa_proc_1.isIn = [hosp_out_gyn_cli_spo_1]
```



Mary (patient)



Dr. Smith (gynecologist)



Manual breast exam (process)



Clinic (location)



Time instant



Plan (reason)



Observation (breast lump)

#1 Medical Procedure & Observation result

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Manual breast exam (process)
-  Clinic (location)
-  Time instant
-  Plan (reason)
-  Observation (breast lump)

```
# the reason why the examination was done

rou_gyn_exa_proc_1 = SCT_RoutineGynecologicExamination("rou_gyn_exa_proc_1")
planned_rou_gyn_qual_1 = SCT_Planned("planned_rou_gyn_qual_1")

# plan to perform a routine gyn exam

plan_rou_gyn_exa_io_1 = sulo.InformationObject("plan_rou_gyn_exa_io_1")
plan_rou_gyn_exa_io_1.refersTo = [rou_gyn_exa_proc_1]
plan_rou_gyn_exa_io_1.hasFeature = [planned_rou_gyn_qual_1]

#the plan has the role "reason" in the manual breast examination

rou_gyn_exa_reason_role_1 = sulo.Role("rou_gyn_exa_reason_role_1")
rou_gyn_exa_reason_role_1.isFeatureOf = [plan_rou_gyn_exa_io_1]
rou_gyn_exa_reason_role_1.isParticipantIn = [man_bre_exa_proc_1]
```

#1 Medical Procedure & Observation result

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Manual breast exam (process)
-  Clinic (location)
-  Time instant
-  Plan (reason)
-  Observation (breast lump)

```
# left breast lump

lef_bre_spo_1 = SCT_LeftBreast("lef_bre_pro_1")
lef_bre_lum_pro_1 = SCT_BreastLump("lef_bre_lum_pro_1")
lef_bre_lum_pro_1.isIn = [lef_bre_spo_1]

# observation made as result of the exam

stmt_lef_bre_lum_io_1 = sulo.InformationObject("stmt_lef_bre_lum_io_1")
stmt_lef_bre_lum_io_1.refersTo = [lef_bre_lum_pro_1]

#the stmt has the role "output" in the manual breast examination

stmt_lef_bre_lum_output_role_1 = sulo.Role("stmt_lef_bre_lum_output_role_1")
stmt_lef_bre_lum_output_role_1.isFeatureOf = [stmt_lef_bre_lum_io_1]
stmt_lef_bre_lum_output_role_1.isParticipantIn = [man_bre_exa_proc_1]
```

#2 Medical Procedure & Observation result (Your turn!!)



On 2026-02-20, an ultrasound of left breast is performed. The radiologist (Dr. Miller) interprets the images and he observes a suspicious mass in the left breast that he documents as a tumor finding.



[Link to notebook #2](#)

Healthcare Data Modelling Tutorial

#3 Diagnosed condition

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Diagnostic assessment (process)
-  2026-02-22 (time instant)
-  Diagnosis (suspected breast cancer – preliminary diagnosis, pending confirmation)

Based on the clinical examination and imaging findings, on the 2026-02-22 Dr. Smith records a suspected breast cancer, as a provisional/ preliminary diagnosis, pending histopathological confirmation.

1. Use SNOMED CT ontology classes whenever possible

a. **SCT_MalignantNeoplasmOfBreast:**

254837009 | Malignant neoplasm of breast (**disorder**) |

a. **SCT_PreliminaryDiagnosis:**

148006 | Preliminary diagnosis (contextual qualifier) (**qualifier value**) |

a. **SCT_DiagnosticAssessment:**

165197003 | Diagnostic assessment (**procedure**) |

Healthcare Data Modelling Tutorial

#3 Diagnosed condition

1. Align SNOMED CT classes with SULO upper-level ones

sulo:Process

a. **SCT_MalignantNeoplasmOfBreast:**

254837009 | Malignant neoplasm of breast (**disorder**) |

sulo:Quality

a. **SCT_PreliminaryDiagnosis:**

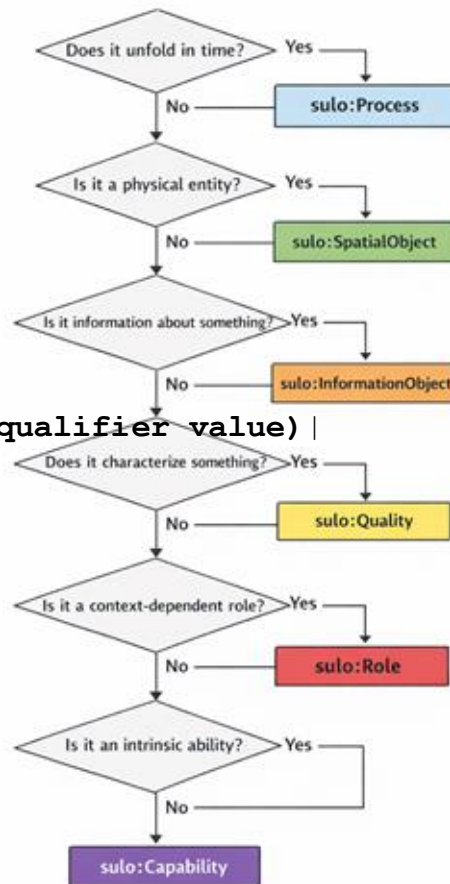
148006 | Preliminary diagnosis (contextual qualifier) (**qualifier value**) |

sulo:Process

a. **SCT_DiagnosticAssessment:**

165197003 | Diagnostic assessment (**procedure**) |

SULO Classification Decision Flowchart



Healthcare Data Modelling Tutorial

#3 Diagnosed condition

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Diagnostic assessment (process)
-  2026-02-22 (time instant)
-  Diagnosis (suspected breast cancer – preliminary diagnosis, pending confirmation)

1. Create the necessary ontology instances for representing the use case

```
# instance of diagnostic process/assessment
diag_proc_1 = SCT_DiagnosticAssessment("diag_proc_1")

# Both, the gynecologist and the patient are process participants
diag_proc_1.hasParticipant = [pat_mary_sulo_role, smith_gyn_role]

# time of the diagnosis
diag_tinst_1 = sulo.TimeInstant("diag_tinst_1")
diag_tinst_1.hasValue = datetime.datetime(2026,2,22,0,0)
diag_proc_1.atTime = [diag_tinst_1]
```

[Link to notebook #3](#)

Healthcare Data Modelling Tutorial

#3 Diagnosed condition

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Diagnostic assessment (process)
-  2026-02-22 (time instant)
-  Diagnosis (suspected breast cancer – preliminary diagnosis, pending confirmation)

```
# instance of the condition
mal_neo_bre_cond_1 = SCT_MalignantNeoplasmOfBreast("mal_neo_bre_cond_1")

# diagnostic statement
stmt_mal_neo_bre_io_1 = sulo.InformationObject("stmt_mal_neo_bre_io_1")
stmt_mal_neo_bre_io_1.refersTo = [mal_neo_bre_cond_1]

# preliminary diagnosis
pre_diag_qual_1 = SCT_PreliminaryDiagnosis("pre_diag_qual_1")
stmt_mal_neo_bre_io_1.hasFeature = [pre_diag_qual_1]
```

#3 Diagnosed condition



Mary (patient)



Dr. Smith (gynecologist)



Diagnostic assessment (process)



2026-02-22 (time instant)



Diagnosis (suspected breast cancer – preliminary diagnosis, pending confirmation)

The statement referring to the neoplasm is the result of the diagnostic assessment

```
stmt_mal_neo_bre_output_role_1 = sulo.Role("stmt_mal_neo_bre_output_role_1")
```

```
stmt_mal_neo_bre_output_role_1.isFeatureOf = [stmt_mal_neo_bre_io_1]
```

```
stmt_mal_neo_bre_output_role_1.isParticipantIn = [diag_proc_1]
```

The statement referring to the tumor is the evidence or reason for this diagnosis

```
stmt_tumor_reason_role_1 = sulo.Role("stmt_tumor_reason_role_1")
```

```
stmt_tumor_reason_role_1.isFeatureOf = [stmt_tumor_io_1]
```

```
stmt_tumor_reason_role_1.isParticipantIn = [diag_proc_1]
```

#4 Medical Procedure & Spatial Object as result

On 2026-02-25, a biopsy of the left breast is performed. A specimen from the left breast obtained by biopsy is produced and submitted for pathological examination.

1. Use SNOMED CT ontology classes whenever possible

a. **SCT_BiopsyOfBreast:**

122548005 | Biopsy of breast (**procedure**) |

a. **SCT_SpecimenBreastObtainedByBiopsy:**

122601001 | Specimen from breast obtained by biopsy (**specimen**) |

-  Mary (patient)
-  Dr. Smith (gynecologist)
-  Biopsy of left breast (process)
-  2026-02-25 (time instant)
-  Result (specimen obtained from the left breast for pathological examination)

#4 Medical Procedure & Spatial Object as result

1. Align SNOMED CT classes with SULO upper-level ones

sulo:Process

a. **SCT_BiopsyOfBreast:**

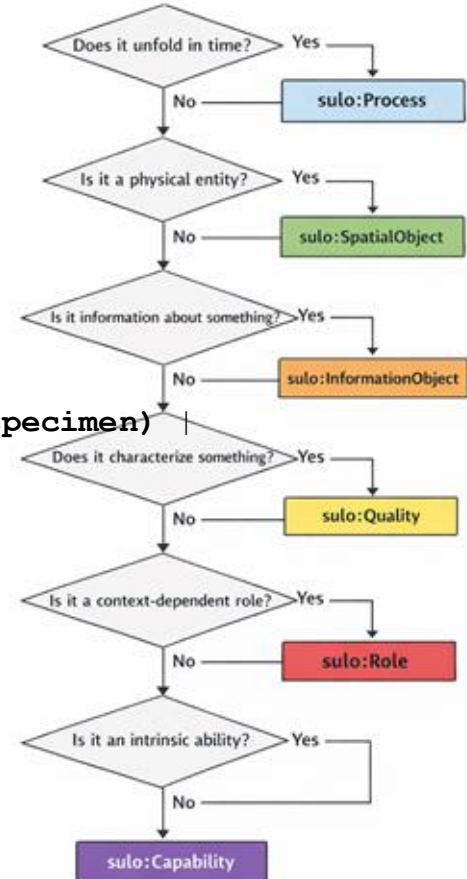
122548005 | Biopsy of breast (**procedure**) |

sulo:SpatialObject

a. **SCT_SpecimenBreastObtainedByBiopsy:**

122601001 | Specimen from breast obtained by biopsy (**specimen**) |

SULO Classification Decision Flowchart



#4 Medical Procedure & Spatial Object as result

1. Create the necessary ontology instances for representing the use case

[Link to notebook #4](#)

```
# instance of the biopsy of left breast
bio_lef_bre_proc_1 = SCT_BiopsyOfBreast("bio_lef_bre_proc_1")
bio_lef_bre_proc_1.hasParticipant = [lef_bre_spo_1]

# Both, the gynecologist and the patient are process participants
bio_lef_bre_proc_1.hasParticipant = [pat_mary_role, gyn_smith_role]

# time of the biopsy
bio_bre_tinst_1 = sulo.TimeInstant("bio_bre_tinst_1")
bio_bre_tinst_1.hasValue = datetime.datetime(2026,2,25,0,0)
bio_lef_bre_proc_1.atTime = [bio_bre_tinst_1]
```

#4 Medical Procedure & Spatial Object as result

 Mary (patient)
 Dr. Smith (gynecologist)
 Biopsy of left breast (process)
 2026-02-25 (time instant)
 Result (specimen obtained from the left breast for pathological examination)

```
# specimen from the breast
spe_bre_by_bio_spo_1 =
    SCT_SpecimenBreastObtainedByBiopsy("spe_bre_by_bio_spo_1")

# the breast specimen is participant in the biopsy, being a result
    spe_bre_by_bio_output_role_1 = sulo.Role("spe_bre_by_bio_output_role_1")
spe_bre_by_bio_output_role_1.isFeatureOf = [spe_bre_by_bio_spo_1]
spe_bre_by_bio_output_role_1.isParticipantIn = [bio_lef_bre_proc_1]
```


#5 Medical Procedure & Measurement results

On 2026-03-01, a pathological analysis was performed. Tumor grade 2 was determined.

Receptor status was also assessed during the pathological analysis.

Estrogen receptor status was positive. Progesterone receptor status was positive. HER2 status was negative.

1. Use SNOMED CT ontology classes whenever possible

a. **SCT_HistopathologyTest:**

252416005 | Histopathology test (**procedure**) |

a. **SCT_HistologicGradeOfNeoplasm:**

371469007 | Histologic grade of neoplasm (**observable entity**) |

a. **SCT_PresenceOfERInNeoplasm:**

445028008 | Presence of estrogen receptor in neoplasm (**observable entity**) |

a. **SCT_PresenceOfPRInNeoplasm:**

45029000 | Presence of progesterone receptor in neoplasm (**observable entity**) |

a. **SCT_PresenceOfHER2InNeoplasm:**

3550001000004108 | Presence of receptor tyrosine-protein kinase erbB-2 in primary malignant neoplasm of breast by immunohistochemistry (**observable entity**) |



Mary (patient)



Breast specimen (participant)



Histopathology test (process)



2026-03-01 (time instant)



Result (tumor grade 2)



Result (estrogen receptor positive)



Result (progesterone receptor positive)



Result (HER2 negative)

#5 Medical Procedure & Measurement results

-  Mary (patient)
-  Breast specimen (participant)
-  Histopathology test (process)
-  2026-03-01 (time instant)
-  Result (tumor grade 2)
-  Result (estrogen receptor positive)
-  Result (progesterone receptor positive)
-  Result (HER2 negative)

1. Align SNOMED CT classes with SULO upper-level ones

sulo:Process

a. **SCT_HistopathologyTest:**

252416005 | Histopathology test (**procedure**) |

sulo:InformationObject

b. **SCT_HistologicGradeOfNeoplasm:**

371469007 | Histologic grade of neoplasm (**observable entity**) |

sulo:InformationObject

c. **SCT_PresenceOfERInNeoplasm:**

445028008 | Presence of estrogen receptor in neoplasm (**observable entity**) |

sulo:InformationObject

d. **SCT_PresenceOfPRInNeoplasm:**

45029000 | Presence of progesterone receptor in neoplasm (**observable entity**) |

sulo:InformationObject

e. **SCT_PresenceOfHER2InNeoplasm:**

3550001000004108 | Presence of receptor tyrosine-protein kinase erbB-2 in primary malignant neoplasm of breast by immunohistochemistry (**observable entity**) |

#5 Medical Procedure & Measurement results

 **Histologic grade of neoplasm (observable entity)**  

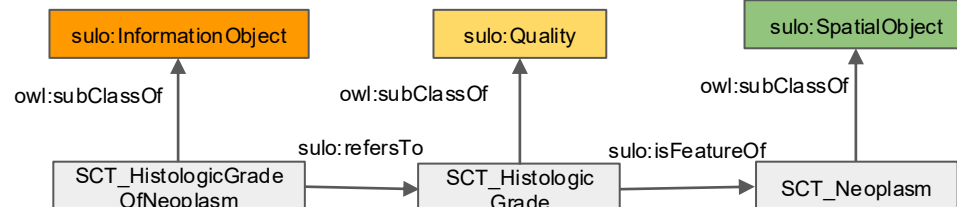
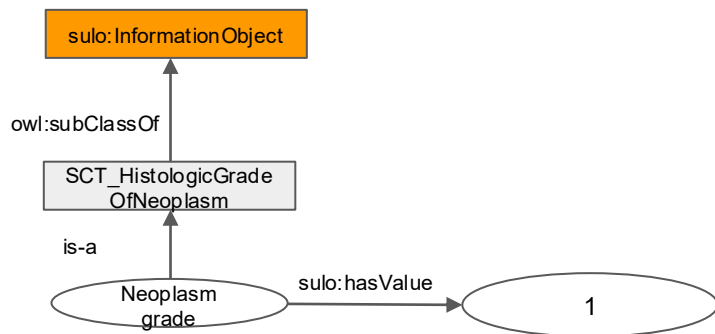
SCTID: 371469007

371469007 | Histologic grade of neoplasm (observable entity) |

- en Histologic grade of neoplasm (observable entity)
- en Histologic grade of neoplasm
- en Histologic grade

Inheres in → Neoplasm
Property → Histologic grade (property)
Scale type → Ordinal value
Time aspect → Single point in time

```
stmt_pre_er_neo_qty_1 = SCT_HistologicGradeOfNeoplasm("stmt_pre_er_neo_qty_1")  
stmt_pre_er_neo_qty_1.hasValue = int(1)
```



#5 Medical Procedure & Measurement results

 **Presence of estrogen receptor in neoplasm**  
(observable entity)
SCTID: 445028008

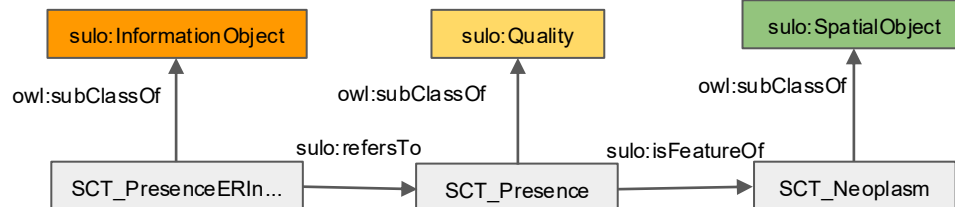
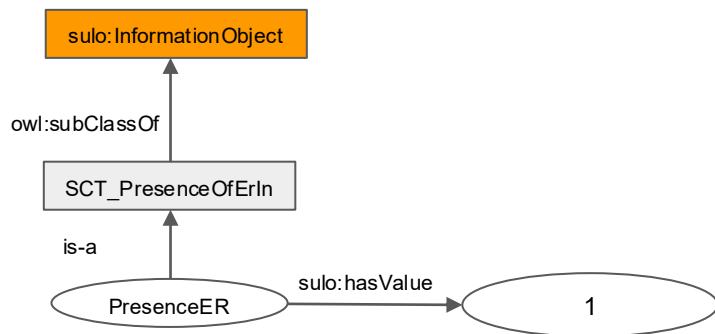
445028008 | Presence of estrogen receptor in neoplasm (observable entity) |

- en Presence of estrogen receptor in neoplasm (observable entity)
- en Presence of estrogen receptor in neoplasm
- en Status of estrogen receptors of neoplasm
- en Presence of oestrogen receptor in neoplasm
- en Status of oestrogen receptors of neoplasm

Inheres in → Neoplasm
Property → Presence (property)
Scale type → Ordinal value
Component → Estrogen receptor
Time aspect → Single point in time

```
stmt_pre_er_neo_qty_1 = SCT_PresenceOfERInNeoplasm("stmt_pre_er_neo_qty_1")
```

```
stmt_pre_er_neo_qty_1.hasValue = int(1)
```



#5 Medical Procedure & Measurement results

-  Mary (patient)
-  Breast specimen (participant)
-  Histopathology test (process)
-  2026-03-01 (time instant)
-  Result (tumor grade 2)
-  Result (estrogen receptor positive)
-  Result (progesterone receptor positive)
-  Result (HER2 negative)

1. Create the necessary ontology instances for representing the use case

```
# instance of the histopathology test on the specimen
his_test_proc_1 = SCT_HistopathologyTest("his_test_proc_1")
his_test_proc_1.hasParticipant = [spe_bre_by_bio_spo_1, pat_mary_role]

# time of the histopathology test
his_test_tinst_1 = sulo.TimeInstant("his_test_tinst_1")
his_test_tinst_1.hasValue = datetime.datetime(2026,3,1,0,0)
his_test_proc_1.atTime = [his_test_tinst_1]
```

[Link to notebook #5](#)

#5 Medical Procedure & Measurement results

-  Mary (patient)
-  Breast specimen (participant)
-  Histopathology test (process)
-  2026-03-01 (time instant)
-  Result (tumor grade 2)
-  Result (estrogen receptor positive)
-  Result (progesterone receptor positive)
-  Result (HER2 negative)

```
# Tumor grade 2

his_gra_neo_io_1 = SCT_HistologicGradeOfNeoplasia("his_gra_neo_io_1")
his_gra_neo_io_1.hasValue = int(2)

# tumor grade is a result of the histopathology test

his_gra_neo_output_role_1 = sulo.Role("his_gra_neo_output_role_1")
his_gra_neo_output_role_1.isFeatureOf = [his_gra_neo_io_1]
his_gra_neo_output_role_1.isParticipantIn = [his_test_proc_1]
```

#5 Medical Procedure & Measurement results

-  Mary (patient)
-  Breast specimen (participant)
-  Histopathology test (process)
-  2026-03-01 (time instant)
-  Result (tumor grade 2)
-  Result (estrogen receptor positive)
-  Result (progesterone receptor positive)
-  Result (HER2 negative)

```
# Lab result ER positive
pre_er_neo_io_1 = SCT_PresenceOfERInNeoplasia("stmt_pre_er_neo_qty_1")
pre_er_neo_io_1.hasValue = int(1)
```

```
# Lab result ER positive is a result of the histopathology test
pre_er_neo_output_role_1 = sulo.Role("pre_er_neo_output_role_1")
pre_er_neo_output_role_1.isFeatureOf = [pre_er_neo_io_1]
pre_er_neo_output_role_1.isParticipantIn = [his_test_proc_1]
```

#5 Medical Procedure & Measurement results

```
# Lab result PR positive

pre_pr_neo_io_1 = SCT_PresenceOfPRInNeoplasia("pre_pr_neo_io_1")
    pre_pr_neo_io_1.hasValue = int(1)

# Lab result PR positive is a result of the histopathology test
pre_pr_neo_output_role_1 = sulo.Role("pre_pr_neo_output_role_1")
pre_pr_neo_output_role_1.isFeatureOf = [pre_pr_neo_io_1]
pre_pr_neo_output_role_1.isParticipantIn = [his_test_proc_1]
```

-  Mary (patient)
-  Breast specimen (participant)
-  Histopathology test (process)
-  2026-03-01 (time instant)
-  Result (tumor grade 2)
-  Result (estrogen receptor positive)
-  Result (progesterone receptor positive)
-  Result (HER2 negative)

#5 Medical Procedure & Measurement results

-  Mary (patient)
-  Breast specimen (participant)
-  Histopathology test (process)
-  2026-03-01 (time instant)
-  Result (tumor grade 2)
-  Result (estrogen receptor positive)
-  Result (progesterone receptor positive)
-  Result (HER2 negative)

```
# Lab result HER2 negative
```

```
pre_her2_neo_io_1 = SCT_PresenceOfHER2InNeoplasia("pre_her2_neo_qty_1")  
    pre_her2_neo_io_1.hasValue = int(0)
```

```
# Lab result HER2 negative is a result of the histopathology test
```

```
pre_her2_neo_output_role_1 = sulo.Role("pre_her2_neo_output_role_1")  
pre_her2_neo_output_role_1.isFeatureOf = [pre_her2_neo_io_1]  
pre_her2_neo_output_role_1.isParticipantIn = [his_test_proc_1]
```

Healthcare Data Modelling Tutorial

#6 Diagnosed condition



The pathological analysis on 2026-03-01 confirms the diagnosis of invasive carcinoma of the left breast.



[Link to notebook #6](#)

Healthcare Data Modelling Tutorial

#7 Medical Treatment



Mary (patient)



Neoadjuvant antineoplastic chemotherapy (process)



2026-03-10 → 2026-06-15 (time interval)



Treatment (AC-T regimen: doxorubicin, cyclophosphamide → paclitaxel)

From 2026-03-10 to 2026-06-15, the patient undergoes neoadjuvant antineoplastic chemotherapy consisting of doxorubicin and cyclophosphamide followed by paclitaxel (AC-T regimen).

1. Use SNOMED CT ontology classes whenever possible

a. **SCT_NeoadjuvantAntineoplasticChemotherapy:**

1259200004 | Neoadjuvant antineoplastic chemotherapy (**procedure**) |

a. **SCT_Doxorubicin:**

372817009 | Doxorubicin (**substance**) |

a. **SCT_Cyclophosphamide:**

387420009 | Cyclophosphamide (**substance**) |

a. **SCT_Paclitaxel:**

387374002 | Paclitaxel (**substance**) |

Healthcare Data Modelling Tutorial

#7 Medical Treatment



Mary (patient)



Neoadjuvant antineoplastic chemotherapy (process)



2026-03-10 → 2026-06-15 (time interval)



Treatment (AC-T regimen: doxorubicin, cyclophosphamide → paclitaxel)

1. Align SNOMED CT classes with SULO upper-level ones

sulo:Process

a. **SCT_NeoadjuvantAntineoplasticChemotherapy:**

1259200004 | Neoadjuvant antineoplastic chemotherapy (**procedure**) |

sulo:SpatialObject

a. **SCT_Doxorubicin:**

372817009 | Doxorubicin (**substance**) |

sulo:SpatialObject

a. **SCT_Cyclophosphamide:**

387420009 | Cyclophosphamide (**substance**) |

sulo:SpatialObject

a. **SCT_Paclitaxel:**

387374002 | Paclitaxel (**substance**) |

Healthcare Data Modelling Tutorial

#7 Medical Treatment



Mary (patient)



Neoadjuvant antineoplastic chemotherapy (process)



2026-03-10 → 2026-06-15 (time interval)



Treatment (AC-T regimen: doxorubicin, cyclophosphamide → paclitaxel)

1. Create the necessary ontology instances for representing the use case

```
# instance of the chemotherapy process
```

```
neo_ant_chem_proc_1 = SCT_NeoadjuvantAntineoplasticChemotherapy("neo_ant_chem_proc_1")
```

```
neo_ant_chem_proc_1.hasParticipant = [pat_mary_role]
```

[Link to notebook #7](#)

```
# start time of the chemotherapy
```

```
neo_ant_chem_stime_1 = sulo.StartTime("neo_ant_chem_stime_1")
```

```
neo_ant_chem_stime_1.hasValue = datetime.datetime(2026,3,10,0,0)
```

```
neo_ant_chem_proc_1.atTime = [neo_ant_chem_stime_1]
```

```
# end time of the chemotherapy
```

```
neo_ant_chem_etime_1 = sulo.EndTime("neo_ant_chem_etime_1")
```

```
neo_ant_chem_etime_1.hasValue = datetime.datetime(2026,6,15,0,0)
```

```
neo_ant_chem_proc_1.atTime = [neo_ant_chem_etime_1]
```

Healthcare Data Modelling Tutorial

#7 Medical Treatment



Mary (patient)



Neoadjuvant antineoplastic chemotherapy (process)



2026-03-10 → 2026-06-15 (time interval)



Treatment (AC-T regimen: doxorubicin, cyclophosphamide → paclitaxel)

```
# instance of doxorubicin
doxo_spo_1 = SCT_Doxorubicin("doxo_spo_1")

# instance of cyclophosphamide
cyphos_spo_1 = SCT_Cyclophosphamide("cyphos_spo_1")

# instance of paclitaxel
pcltxl_spo_1 = SCT_Paclitaxel("pcltxl_spo_1")

# The three of them have an agent role in the corresponding chemo process
doxo_agent_role_1 = sulo.Role("doxo_agent_role_1")
doxo_agent_role_1.isFeatureOf = [doxo_spo_1]

cyphos_agent_role_1 = sulo.Role("cyphos_agent_role_1")
cyphos_agent_role_1.isFeatureOf = [cyphos_spo_1]

pcltxl_agent_role_1 = sulo.Role("pcltxl_agent_role_1")
pcltxl_agent_role_1.isFeatureOf = [pcltxl_spo_1]
```

Healthcare Data Modelling Tutorial

#7 Medical Treatment



Mary (patient)



Neoadjuvant antineoplastic chemotherapy (process)



2026-03-10 → 2026-06-15 (time interval)



Treatment (AC-T regimen: doxorubicin, cyclophosphamide → paclitaxel)

```
#AC regime
neo_ant_chem_acreg_proc_1 =
    SCT_NeoadjuvantAntineoplasticChemotherapy("neo_ant_chem_acreg_proc_1")
neo_ant_chem_acreg_proc_1.hasParticipant = [pat_mary_role, doxo_agent_role_1, cyphos_agent_role_1]

#T regime
neo_ant_chem_treg_proc_1 =
    SCT_NeoadjuvantAntineoplasticChemotherapy("neo_ant_chem_treg_proc_1")
neo_ant_chem_treg_proc_1.hasParticipant = [pat_mary_role, pcltxl_agent_role_1]

#AC-T regime (First AC, later T)
neo_ant_chem_acreg_proc_1.precedes = [neo_ant_chem_treg_proc_1]

#AC-T processes are part of the main chemo process
neo_ant_chem_proc_1.hasPart = [neo_ant_chem_acreg_proc_1, neo_ant_chem_treg_proc_1]
```

Healthcare Data Modelling Tutorial

#8 Medical Treatment



On 2026-07-01, surgical treatment is performed in the form of lumpectomy of the left breast



[Link to notebook #8](#)

Healthcare Data Modelling Tutorial

#9 Medical Procedure & Diagnosis

On 2026-09-30, follow-up assessment indicates remission.



[Link to notebook #9](#)

Healthcare Data Modelling Tutorial

#9 Medical Procedure & Diagnosis

 Mary (patient)
 Dr. Smith (gynecologist)
 Follow-up visit (process)
 2026-09-30 (time instant)
 Diagnosis (disease in remission)

```
# instance of the condition
inv_car_lef_bre_cond_2 = SCT_InvasiveCarcinomaOfBreast("inv_car_lef_bre_cond_2")
    inv_car_lef_bre_cond_2.isIn = [lef_bre_spo_1]

# diagnostic statement

    stmt_inv_car_lef_bre_io_2 = sulo.InformationObject("stmt_inv_car_lef_bre_io_2")
stmt_inv_car_lef_bre_io_2.refersTo = [inv_car_lef_bre_cond_2]

# output role of the diagnostic statement
stmt_inv_car_lef_bre_output_role_2 = sulo.Role("stmt_inv_car_lef_bre_output_role_2")
stmt_inv_car_lef_bre_output_role_2.isFeatureOf = [stmt_inv_car_lef_bre_io_2]
stmt_inv_car_lef_bre_output_role_2.isParticipantIn = [diag_proc_3]

# disease in remission
rem_phase_qual_1 = SCT_RemissionPhase("rem_phase_qual_1")
inv_car_lef_bre_cond_2.hasFeature = [rem_phase_qual_1]
```

Healthcare Data Modelling Tutorial

#10 Planned Procedure



Mary (patient)



Dr. Smith (gynecologist)



Follow-up visit (planned process)



2027-01-01 (scheduled time)

A follow-up visit is scheduled for 01/01/2027.

1. Use SNOMED CT ontology classes whenever possible

a. **SCT_FollowUpVisit:**

185389009 | Follow-up visit (**procedure**) |

Healthcare Data Modelling Tutorial

#10 Planned Procedure

 Mary (patient)
 Dr. Smith (gynecologist)
 Follow-up visit (planned process)
 2027-01-01 (scheduled time)

1. Align SNOMED CT classes with SULO upper-level ones

sulo:Process

a. **SCT_FollowUpVisit:**

185389009 | Follow-up visit (**procedure**) |

Healthcare Data Modelling Tutorial

#10 Planned Procedure

[Link to notebook #10](#)



Mary (patient)



Dr. Smith (gynecologist)



Follow-up visit (planned process)



2027-01-01 (scheduled time)

1. Create the necessary ontology instances for representing the use case

```
# instance of the follow-up visit
```

```
follow_up_proc_2 = SCT_FollowUpVisit("follow_up_proc_2")
```

```
follow_up_proc_2.hasParticipant = [pat_mary_role, gyn_smith_role]
```

```
# date time for the follow-up
```

```
follow_up_tinst_2 = sulo.StartTime("follow_up_tinst_1 ")
```

```
follow_up_tinst_2.hasValue = datetime.datetime(2027,1,1,0,0)
```

```
# plan for the follow-up
```

```
planstmt_follow_up_io_2 = sulo.InformationObject("planstmt_follow_up_io_2")
```

```
planstmt_follow_up_io_2.refersTo = [follow_up_proc_2]
```

```
start_process_plan_io_2 = sulo.InformationObject("start_process_plan_io_2")
```

```
    planstmt_follow_up_io_2.hasPart = [start_process_plan_io_2]
```

```
start_process_plan_io_2.hasPart = [follow_up_tinst_2]
```